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data . . .
1 2 3 4 5 6

Col 20382

- 1 - a - $\{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30\}$
b - $\{1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29\}$
c - $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$
d - $\{17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$
e - $\{10, 20, 30\}$
f - $\{3, 6, 8, 9, 12, 15, 16, 18, 21, 24, 27, 30\}$
g - $\{6, 12, 18, 24, 30\}$

- 2 - a - $\{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)\}$
b - $\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$
c - $\{(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$
d - $\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$
e - $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$

- 4 - a - $\{(1, 1), (2, 2), (3, 3), (3, 4)\}$
b - $\{(2, 1), (3, 1), (4, 1), (3, 2), (4, 2), (4, 3)\}$
c - $\{(1, 1)\}$
d - $\{(1, 1), (2, 4)\}$
e - $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5)\}$
f - $\{(1, 3), (2, 3), (3, 3), (4, 3)\}$

- 5 - a - $\{(1, V), (1, B)\}$
b - $\{(11, V), (11, B)\}$
c - $\{(1, V), (11, B)\}$
d - $\{(1, B), (11, B)\}$

$$4 - a - A \times B = \{(1,1), (1,2), (1,3), (1,4), (1,5), (2,1), (2,2), (2,3), (2,4), (2,5), (3,1), (3,2), (3,3), (3,4), (3,5), (4,1), (4,2), (4,3), (4,4), (4,5)\}$$

$$A = \{(1,1), (2,2), (3,3), (4,4)\}$$

$$b - B = \{(2,3), (2,4), (2,5), (3,4), (3,5), (4,5)\}$$

$$c - C = \{(1,1)\}$$

$$d - D = \{(1,1), (1,4)\}$$

$$e - E = \{(1,1), (1,2), (1,3), (1,4), (1,5)\}$$

$$f - F = \{(1,3), (2,3), (3,3), (4,3)\}$$

$$6 - a - S = \{0, 1, 2, 3, 4, \dots\}$$

$$A = \{12, 13, 14, \dots\} = \{x \mid x > 11\}$$

$$B = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$$

$$= \{x \mid x \leq 15\}$$

$$C = \{8, 9, 10, 11\} = \{x \mid 8 \leq x < 12\}$$

$$b - A \cup B = \{0, 1, 2, 3, 4, \dots\}$$

$$c - A \cap B = \{12, 13, 14, 15\}$$

$$d - \bar{A} = \{0, 1, 2, 3, \dots, 11\}$$

$$e - A \cup B \cup C = \{0, 1, 2, 3, 4, \dots\}$$

$$f - (A \cup C)' = \{0, 1, 2, 3, 4, \dots\}$$

$$g - A \cap B \cap C = \emptyset$$

$$h. B' \cap C = \emptyset$$

$$i). A \cup (B \cap C) = \{8, 9, 10, 11, 12, \dots\}$$

$$7. a. 2 \cdot 2 \cdot 2 \cdot 2 = 16 \text{ elementos}$$

$$E = \{(d d d d), (d o d d), (o d d d), (o o d d), (d d d o), (d o d o), (o d d o), (o o d o), (d d o o), (d o o o), (o d o o), (o o o o)\}$$

$$b. A_1 = \{(d d d d), (d o d d), (d o d o), (d o d o), (d d o d), (d o o d), (d d o o), (d o o o)\}$$

$$c. A'_1 = \{(o d d d), (o o d d), (o d d o), (o d o d), (o o d o), (o o d o), (o d o o), (o o o o)\}$$

$$d. A_1 \cap A_2 \cap A_3 \cap A_4 = \{(d d d d)\}$$

$$e. (A_1 \cap A_2) \cup (A_3 \cap A_4) = \{(d d d d), (d d d o), (d d o d), (d d o o), (d o d d), (o d d d), (o o d d), (o o d d)\}$$

$$8. a. A' \cap B = 55$$

$$A' = \{25, 30, 5, 10\}$$

$$B = \{22, 25, 30\}$$

$$A' \cap B = \{25 + 30 + 5 + 10\} + \{22 + 25 + 30\} - (25 + 30 + 10 + 22) = (70 + 77) - 92 = 147 - 92 = 55$$

$$b. B' = \{8, 5, 10\}$$

$$= (8 + 5 + 10)$$

$$= 23$$

$$c - A \cup B = (22+8) + (22+25+30) - 22$$

$$A = \{22, 8\} = (30+77) - 22$$

$$= 307 - 22$$

$$= 85$$

$$g - 2.2.2 = 8 \mid E = \{(ppp), (nnn), (ppn), (pnp), (pnn), (npp), (nnp), (nnp)\}$$

$$g - a - A = \{(ppp)\}$$

$$b - B = \{(nnn)\}$$

$$c - A \cap B = \emptyset$$

$$d - A \cup B = \{(ppp), (nnn)\}$$

$$10 - a) A \cap B = 570 + 443 + 60 = 673$$

$$b) A' = 28 + 363 + 309 + 933 + 39 = 1672$$

$$c - A \cup B = 6823 + 767 - 673 = 6915$$

$$d - A \cup B' = 6823 + 0 = 6823$$

$$e - A' \cap B' = \emptyset$$